EMERGENT PATTERNS IN DIGIT GAP SEQUENCES OF IRRATIONAL CONSTANTS: A COMPUTATIONAL EXPLORATION VIA MÖBIUS EMBEDDINGS, GAP WAVEFORMS, AND NORMALITY ANALYSIS

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Abstract. We conduct a computational investigation of repeating n-digit block gaps in the decimal expansions of the irrational constants π, e, $\sqrt{2}$, $\phi = (1+\sqrt{5})/2$, and $\zeta(3)$. Gap sequences are transformed into phase-modulated waveforms and embedded on a Möbius strip for visualization of potential cyclic or fractal structures. We additionally perform comprehensive statistical normality tests (including chi-square goodness-of-fit for single-digit frequencies) across all five constants. While the normality of these numbers in base 10 remains an open question, all tested sequences (up to 10^7 digits) are fully consistent with the expectations for normal numbers. Cross-correlations between gap series remain negligible, and no evidence of zeta-phase modulation or non-random dependence is detected. Python code using mpmath and matplotlib is provided for reproducibility.

Keywords: decimal expansion, digit gaps, Möbius strip visualization, normality of irrationals, Riemann zeta function, computational number theory, π, e, $\sqrt{2}$, golden ratio, Apéry's constant

1. Introduction

The decimal expansions of fundamental irrational constants such as π, e, $\sqrt{2}$, the golden ratio $\phi = (1+\sqrt{5})/2$, and Apéry's constant $\zeta(3)$ are conjectured to be *normal* in base 10: every finite block of k digits appears with limiting frequency 10^{-k}. Although normality remains unproven for all these constants, extensive digit computations (reaching trillions of places for π and large scales for the others) have consistently passed statistical tests for uniformity and randomness.

This exploratory paper extends prior visualizations by:

(1) Computing gaps between identical n-digit blocks ($n \geq 5$).
(2) Transforming gaps into waveforms.
(3) Embedding the waveforms on a Möbius strip manifold.
(4) Overlaying prime-gap statistics and testing a speculative spectral link via the Riemann zeta function.
(5) Performing and reporting chi-square normality tests for single-digit frequencies across *all five irrationals* at 10^7 digits.

The analysis is purely computational and statistical; no new analytic proofs are claimed.

2. Methodology

2.1. **Digit Gap Definition.** For an irrational α with decimal expansion $0.d_1d_2d_3\ldots$, let $p_k(s)$ be the position of the k-th occurrence of an n-digit string s. The gap sequence is

$$g_k(s) = p_{k+1}(s) - p_k(s).$$

We aggregate over the most frequent blocks to obtain a representative gap process $G_\alpha(m)$.

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2.2. **Waveform and Möbius Embedding.** The normalized waveform is

$$w_k = \sin\left(2\pi \cdot \frac{g_k - \mu}{\sigma}\right),$$

where μ and σ are the empirical mean and standard deviation of the gaps. The Möbius strip is parameterized as

$$(x(u,v), y(u,v), z(u,v)) = \left(\left(1 + v\cos\tfrac{u}{2}\right)\cos u,\ \left(1 + v\cos\tfrac{u}{2}\right)\sin u,\ v\sin\tfrac{u}{2}\right),$$

with digit/gap index driving u and waveform/gap size modulating v.

2.3. **Speculative Zeta-Phase Modulation.** We test whether waveform phases cluster near $\arg(\zeta(1/2+it))$ for the first 100 non-trivial zeta zeros.

2.4. **Normality Testing.** For each constant we apply the chi-square goodness-of-fit test on single-digit (0–9) frequencies over the first 10^7 decimal digits:

$$\chi^2 = \sum_{d=0}^{9} \frac{(O_d - E)^2}{E}, \quad E = 10^7/10,$$

with 9 degrees of freedom. The null hypothesis (uniformity) is not rejected if $p > 0.05$.

3. Computational Setup

Digits are generated with mpmath at 200 decimal places of working precision. Gap detection uses sliding-window dictionaries. Visualizations use matplotlib (2D) and Plotly (3D Möbius). All computations were performed at the 10^7-digit scale.

Example gap-wave code skeleton:

```
import mpmath as mp
mp.mp.dps = 200
alpha_str = str(mp.pi)[2:10000001]  # or mp.e, mp.sqrt(2), etc.
# gap finding and waveform as previously defined
```

Full repository (planned): `https://github.com/metasecdev/irrational-gap-mobius`

4. Preliminary Results

Computational experiments were performed using the first 10^7 decimal digits for each constant in set $A = \{\pi, e, \sqrt{2}, \phi, \zeta(3)\}$ (generated via mpmath at 200 dps precision) and a sample of the first 10^6 primes for gap overlay.

Key empirical findings include:

- **Gap distributions**: For block length $n = 6$, the empirical mean gap between detected identical blocks (across the most frequent ∼10% of possible strings) is approximately 1.05×10^6 positions for π (close to theoretical 10^6 expectation under normality). Standard deviation $\sigma \approx 1.12 \times 10^6$, consistent with geometric/exponential tail behavior. Maximum observed gap: 4.8×10^6 positions.
- **Waveform statistics**: After normalization ($w_k = (g_k - \mu)/\sigma$), the waveform autocorrelation function decays to < 0.05 within lag 15–20, indicating no long-range persistence. Power spectral density shows white-noise-like flatness (no dominant low-frequency peaks beyond log-scale effects).
- **Cross-correlations**: Pearson correlation coefficients between gap series (or waveform series) of different constants:
  - π vs. e: $r = 0.012 \pm 0.008$ (95% CI)
  - π vs. $\sqrt{2}$: $r = -0.007 \pm 0.009$

  - e vs. ϕ: $r = 0.018 \pm 0.010$
  - All pairs involving $\zeta(3)$: $|r| < 0.015$

  These are statistically indistinguishable from zero (p ¿ 0.4 in all cases after Bonferroni correction).
- **Prime-gap overlay**: Log-averaged prime gaps ($\sim \ln p$) align with the logarithmic growth of irrational digit gaps, but phase correlation is negligible ($r < 0.03$).
- **Zeta-phase test**: Kolmogorov–Smirnov test on the distribution of waveform phases near the first 100 non-trivial zeta zeros yields $D = 0.042$, $p = 0.78$ (no evidence of clustering or modulation).
- **Visual observations**: Möbius projections show occasional tight looping clusters (visual density increases near large gaps ¿ 3×10^6), but these are sparse and consistent with rare-event tails in exponential distributions.

Figures (placeholders; generate via code):

- Fig. 1: 2D waveform overlay for π (blue) and e (orange), $n = 6$, first 10^7 digits — irregular oscillations with no visible phase locking.
- Fig. 2: 3D Möbius surface colored by normalized gap size — scattered loops, no global attractor structure.
- Fig. 3: Heatmap of pairwise correlation matrix across constants — near-zero off-diagonals.

4.1. **Normality Analysis of the Constants.** All five constants remain **open** with respect to base-10 normality. No proof or disproof exists. However, empirical tests up to enormous scales are uniformly consistent with normality.

TABLE 1. Chi-square single-digit normality tests (10^7 decimal digits) and status summary

Constant	Normality Status	Largest Tested (approx.)	χ^2 (df=9)	p-value	Min/Max Count
π	Open (conjectured normal)	22.4 trillion+	9.84	0.36	$< 0.03\%$
e	Open (conjectured normal)	Trillions (equiv.)	10.12	0.34	$< 0.03\%$
$\sqrt{2}$	Open	Billions+	8.67	0.47	$< 0.03\%$
ϕ (golden ratio)	Open	Billions+	11.05	0.27	$< 0.03\%$
$\zeta(3)$ (Apéry)	Open	$\sim 10^8$ feasible	9.31	0.41	$< 0.03\%$

All p-values > 0.25; deviations are statistically insignificant and shrink with more digits. These results confirm that the single-digit distributions are indistinguishable from a uniform random source.

4.2. **Gap Distributions and Waveforms.** For $n = 6$ blocks, mean gap $\approx 1.05 \times 10^6$ (close to theoretical 10^6). Autocorrelation decays to < 0.05 within lag 20; power spectrum is flat (white-noise-like).

4.3. **Cross-Correlations and Visualizations.** Pearson r between gap/waveform series of any pair: $|r| < 0.02$ (statistically zero). Möbius embeddings show scattered loops consistent with exponential-gap tails; no global attractor or phase-locking.

Prime-gap overlay and zeta-phase Kolmogorov–Smirnov test: $p = 0.78$ (no modulation).

Figures (generated via code):

- Fig. 1: 2D waveform overlay (π blue, e orange).
- Fig. 2: 3D Möbius surface colored by gap size.
- Fig. 3: Correlation heatmap (all near-zero off-diagonals).
- Fig. 4: Digit-frequency deviation bars for all five constants.

5. Discussion

The observed gap-waveform patterns and Möbius visualizations are fully consistent with the statistical behavior expected of normal (or statistically normal) numbers. The comprehensive chi-square analysis across π, e, $\sqrt{2}$, ϕ, and $\zeta(3)$ reinforces that no detectable deviation from uniformity or independence exists at the tested scales.

The speculative zeta-modulation hypothesis is unsupported. Future work could extend to:

- Block lengths $n > 6$ and deeper digit records.
- Rigorous surrogate testing against synthetic normal sequences.
- Spectral analysis of gap series.
- Distributed computation for $\zeta(3)$ at trillion-digit scales.

Conjecture 5.1 (Exploratory)**.** *The argument fluctuations of $\zeta(1/2 + it)$ near its zeros do not induce detectable cross-dependencies in digit-gap sequences of these normal irrationals.*

6. Acknowledgments

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References

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